

“Graphs in Statistical Analysis” by **F. J. Anscombe** highlights the importance of graphs in statistical analysis. He highlights that researchers rely too much on performing intricate calculations and skip examining data through graphs. Graphs serve various purposes. It not only allows us to perceive and appreciate the broad features of the data but also let us look behind those broad features. A statistical result may appear true and convincing, but it can sometimes hide important patterns or problems in the data. Most statistical calculations rest on assumptions, and we should always check whether the assumptions are reasonably correct for our data and, if not, we should try to understand in what ways they are incorrect. He demonstrates that statistical results alone cannot provide a complete understanding of a dataset using examples of **regression results**. He suggested that researchers should make graphical analysis an important part of statistical analysis rather than treating graphs as secondary. Before starting the regression analysis, a scatterplot of dependent (y values) and independent values should be prepared to see the sort of relation between the variables. Before conducting regression analysis, it is necessary to determine whether the relationship between the variables appears linear, curved, weak, or affected by unusual observations. If there are “outliers”, then we should check our values again, and if they are correct, then we should set aside the outliers for a special study and fit the regression for the remaining data. If the observation is genuine, its influence on the statistical results should be examined.

Anscombe demonstrates this through **four datasets** that have almost identical numerical summaries, including similar means, variances, correlation coefficients, and regression results, even though the datasets have very different graphical patterns. The first dataset shows a reasonably clear linear relationship, while the second contains a curved relationship, the third is strongly influenced by an outlier, and the fourth dataset shows that the regression relationship is primarily determined by a single outlying observation. For the second dataset, a nonlinear model or transformation of data may be needed, while for the third and fourth datasets, outliers and influential observations should be checked to see whether they are valid or errors. Through this example he demonstrates that statistical results alone cannot provide a complete understanding of a dataset and can hide key features such as nonlinearity, outliers, and influential observations. He also points out that an ordinary least-squares regression assumes that the data follow a linear model with errors drawn from a normal probability distribution having zero mean and constant variance. The calculation gives fitted values and residuals. One should not assume that the regression model is correct just because the calculation is complete. Therefore, we should plot the residuals against the fitted values to reveal patterns, unusual observations, or changing variation. For example, a curved pattern may suggest nonlinearity or unequal spread of residuals, indicating non-constant variance. Then, if there is an outlying x-value, it will affect the regression coefficient and make the regression relation misleading. Therefore, one should also examine the residual plots before accepting the regression results. These effects should be removed so that our regression relation is not misleading.

He also shows how these same principles are also important for regression analysis for multiple independent variables and two-way tables. When several variables are involved, he suggests that researchers should use appropriate graphical techniques, including ordinary scatterplots for every pair of variables, triple scatterplots, graphs of residuals vs. fitted values, and other displays that can reveal relationships between variables. Then, to study the dependence of y on one of the independent variables (x_1), he suggests controlling for another variable (x_2) by removing its contribution from y to examine the relationship between x_1 and y more clearly. Thus, graphical examination will help to identify patterns that might otherwise remain hidden in a table of statistical results.

Therefore, the key learning is that statistical calculations and graphical analysis should be used together. Calculations provide statistical results, while graphs can help understand the structure, relationships, unusual observations, and possible problems within the data. Statistical analysis should, therefore, involve several steps, such as examining the data visually through graphs, performing statistical calculations, checking underlying assumptions through graphs, examining unusual observations, and finally interpreting both the statistical and graphical evidence.